



Dynamic magnetic response of multicore particles: The role of grain magnetic anisotropy and intergrain interactions

Elena S. Pyanzina^a, Ekaterina V. Novak^a, Andrey A. Kuznetsov^b, Sofia S. Kantorovich^{b, ,*}

^a Institute of Natural Sciences and Mathematics, Ural Federal University, Lenin av. 51, 620000, Ekaterinburg, Russia

^b University of Vienna, Faculty of Physics, Kollingasse 14-16, 1090 Vienna, Austria

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ABSTRACT

This study examines the magnetic properties of multicore magnetic nanoparticles (MMNPs) using Brownian dynamics simulations in conjunction with the Landau-Lifshitz-Gilbert equation. We demonstrate that the magnetic responses of MMNPs, both static and dynamic, are governed by a complex interplay between the cores magnetic anisotropy and the strength of intercore magnetic interactions. We show that for a single multicore in a liquid carrier, the distribution of the anisotropy axes inside does not impact its initial magnetic response. These insights are crucial for refining the design of multicore magnetic particles in various applications, including magnetic hyperthermia, data storage, and targeted drug delivery, where accurate control over magnetic characteristics is vital.

1. Introduction

In the last ten years, smart nano-textured materials that can be controlled and manipulated with external stimuli form the basis for novel bio-medical applications. One of the main requirements in this field is that the stimuli are not interfering with any biological or physiological processes. Hence, magnetic fields are optimal to this regard [1], and magnetic soft matter systems have gained a lot of attention for their potential in targeted drug delivery [2–8] and magnetic hyperthermia [9–12].

The beginning of the magnetic soft matter dates back to the synthesis of ferrofluids in the middle of the twentieth century [13]. Today, ferrofluids, also known as magnetic fluids, are considered, perhaps, the simplest example of soft materials that can be controlled magnetically, though they are not fully comprehended and utilised. These complex fluids consist of magnetic single-domain nanoparticles suspended in non-magnetic liquid carriers. The difficulty to precisely manipulate ferrofluids arises from the polydisperse size of magnetic nanoparticles, the specific method of their colloidal stabilisation, and the resulting wide range of time, length, and energy scales involved. In fact, any structures that these nanoparticles form in ferrofluids are highly sensitive to thermal fluctuations and mechanical disturbances. As an alternative approach, recent investigations have suggested replacing single-domain magnetic particles with preassembled complex entities, such as supramolecular polymer-like structures composed of magnetic monomers [14–24], mag-

netic nano- and micro-gels [4,25,26], or magnetic multicore nanoparticles (MMNPs) [27–30].

The latter is the subject of our investigation. The reason for that is their wide range of perspective applications [31]. For instance, Gallo et al. [32] discussed the advantage of multicore particles as self-heating catalysts for the degradation of anionic and cationic dyes by applying magnetic field. The MMNP suspensions have recently been termed “bio-ferrofluids” [33], although whether or not this name is justified is yet to be fully understood. However, MMNPs have already emerged as a source of innovation in the field of drug targeting [34,35], particle imaging [27,36], magnetic hyperthermia cancer treatment [37–40] and immunoassays/biosensing [41–44]. Besides biomedical applications, MMNPs also find use in nanorheology [45–48] and for degradation of the endocrine disrupting chemicals such as bisphenol A in water [49]. A review [50] emphasises that, “among the remotely controlled endogenous ... or exogenous ... stimuli-responsive nanoassemblies, designed to ensure dosage, spatial, and temporal controllability, the magnetic field driven bio-nanocomposites attracted tremendous scientific and technological interest...”. In all these studies, individual MMNPs are considered as clusters of single-domain magnetic nanocrystals (addressed as grains below) embedded in a polymer or other non-magnetic rigid matrix [51,52]. While the grain sizes are typically around 10 nm, the sizes of MMNPs can vary from tens to several hundred nanometers.

Such a difference in characteristic length-scales within MMNP suspensions is manifested in various relaxation scenarios, ranging from

* Corresponding author.

E-mail address: sofia.kantorovich@univie.ac.at (S.S. Kantorovich).

Brownian rotational diffusion of the entire MMNP to internal flip of magnetic moment within individual grains. Predicting and explaining the dominance of one relaxation mechanism over another is crucial for optimising MMNPs. For example, Bertuit et al. [53] showed that the MMNP architecture affects not only hyperthermia, but also photothermia. Bender et al.'s work [54] experimentally investigated the influence of spherical MMNP architecture on magnetic response, revealing contributions to dynamic spectra of MMNP rotation as a whole (Brownian mechanism), internal relaxation of effective magnetisation of MMNP as a whole (Néel-type mechanism at the level of the entire MMNP), and internal relaxation within individual grains (Néel-like mechanism at the level of individual building blocks). The same authors conducted studies on MMNP powders for a more detailed analysis of the latter two relaxation mechanisms [55]. Overall, the architecture and magnetic properties of MMNPs are closely linked to their synthesis process [56,57]. Tomography [58], scattering experiments [59,60], and the FORC (first order reversal curves) diagram method [61–63], among others, are of significant aid for a detailed analysis of internal structure.

It has recently been shown that changes in the shape and architecture of multicore magnetic nanoparticles (MMNPs) significantly affect their thermomagnetic response, opening up broad prospects for various applications [31,64]. Examples of shape modification include ring-shaped [65] and cubic [66,67] MMNPs.

Despite the promising range of potential applications and diverse synthesis possibilities, the key interaction mechanisms in MMNP suspensions, both intrinsic and induced by a magnetic field, remain poorly understood. For example, experiments indicated the differences in the self-assembly of magnetic nanoparticles and complex multicores [68]. Theoretical works describing the influence of magnetic interactions between grains on the equilibrium [69–73] and magnetodynamic response of MMNPs [74–76] have identified similarities and differences with ferrofluid emulsions [77,78], but the processes of MMNP aggregation and their impact on thermomagnetic properties remain unclear.

Questions concerning the impact of shape and architecture on the self-assembly of multicore magnetic particles become even more fundamental in the context of their similarity to colloids with multiple fixed [79–83] and mobile [84–86] patches.

To address these challenges, the initial step involves accounting for the competition between the Brownian relaxation of multicore particles and the magnetic relaxations specific to each core, which are influenced by anisotropy. This study utilizes Brownian dynamics simulations alongside the Landau-Lifshitz-Gilbert equation to examine the static and dynamic magnetic responses of individual MMNPs. We analyze two MMNP configurations: one with randomly oriented anisotropy axes of cores, and another with axes aligned parallel. Initially, the manuscript presents the model and methods, highlighting the integrated Brownian-Landau-Lifshitz-Gilbert simulation technique. Subsequently, we interpret the findings by exploring zero-field dynamic susceptibility spectra and the orientation distributions of core dipoles relative to their respective easy axes, influenced by anisotropy and intercore magnetic interaction strengths. We further evaluate static susceptibilities and perform calculations of the specific loss power. The manuscript concludes with a succinct summary.

2. Model and methods

2.1. Multicore particle

We investigate an isolated spherical multicore particle of diameter D containing N magnetic grains (cores) with diameters d and volumes $v_g = \pi d^3/6$, see Fig. 1. Grains are distributed inside the particle randomly and uniformly, without overlapping. The relative position of the grain within the particle cannot change once assigned. The grain volume fraction within the MMNP is

$$\phi_g = N \frac{v_g}{v_p} = N \left(\frac{d}{D} \right)^3, \quad (1)$$

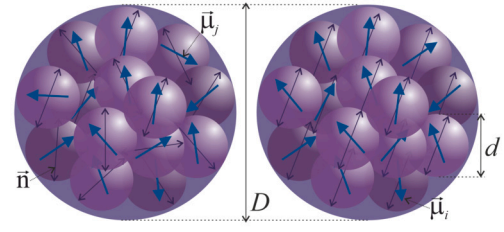


Fig. 1. Sketch of the investigated system. D is diameter of multicore particle, d is grain's diameter, $\vec{\mu}_i$ and $\vec{\mu}_j$ are the dipole moments of grains i and j , and $\vec{n}$ is unit vector of the direction of the easy magnetization axis. On the left each grain has a randomly directed axis of easy magnetization, on the right side all axes are parallel to each other.

where $v_p = \pi D^3/6$ is the overall hydrodynamic volume of the particle. The particle is suspended in a viscous non-magnetic liquid of viscosity η and is thermostated at a constant temperature T .

Each grain i has a magnetic moment $\vec{\mu}_i$ whose magnitude is fixed, $|\vec{\mu}_i| = \mu$, but whose orientation is determined by the interplay between magnetic fields acting on a grain and thermal fluctuations (as described in the dedicated subsection below).

Each grain possesses a uniaxial magnetic anisotropy with an easy axis $\vec{n}_i$. Coupling between the axis and the magnetic moment is given by the potential

$$U_{ani}(i) = -K v_g (\vec{e}_i \cdot \vec{n}_i)^2, \quad (2)$$

where K is anisotropy constant of the grain material, $\vec{e}_i = \vec{\mu}_i/\mu$ is the magnetic moment unit vector. To quantitatively characterize an interplay between anisotropy effects and fluctuations, we will use the dimensionless anisotropy parameter

$$\kappa = \frac{K v_g}{k_B T}, \quad (3)$$

where k_B is the Boltzmann constant.

Each pair of grains interacts via the magnetic dipole-dipole potential

$$U_{dd}(i, j) = \frac{\mu_0}{4\pi} \left[\frac{(\vec{\mu}_i \cdot \vec{\mu}_j)}{r_{ij}^3} - \frac{3(\vec{\mu}_i \cdot \vec{r}_{ij})(\vec{\mu}_j \cdot \vec{r}_{ij})}{r_{ij}^5} \right], \quad (4)$$

where $\vec{r}_{ij}$ is the vector connecting their centres. Corresponding dimensionless parameter is the so-called dipolar coupling constant

$$\lambda = \frac{\mu_0}{4\pi} \frac{\mu^2}{d^3 k_B T}, \quad (5)$$

that is the energy of two adjacent grains whose dipoles are aligned head-to-tail divided by $k_B T$.

Throughout the work, we will focus on multicore particles with $N = 100$ grains and their volume fraction $\phi_g = 0.2$. It corresponds to the diameter $D \simeq 7.94d$. Dipolar coupling constant and anisotropy parameter will be varied in broad ranges: $1 \leq \lambda \leq 10$, $1 \leq \kappa \leq 10$. We will also compare two kinds of particles – *textured multicores*, with easy axes of all grains co-aligned at all times, and *non-textured ones*, with easy axes having a random uniform orientational distribution.

2.2. Equations of motion

To simulate coupled rotational dynamics of multicore particle and magnetic moments of the grains inside of it, we will employ the so-called “Egg model”. It was introduced by Shliomis and Stepanov to describe non-equilibrium magnetic response of dilute ferrofluids [87]. In recent years, it was actively used to model various types of superparamagnetic systems [88–91], including MMNPs [76].

Within the model, the orientation of each easy axis $\vec{n}_i$ is described by the kinematic relationship

$$\frac{d\vec{n}_i}{dt} = \vec{\omega} \times \vec{n}_i. \quad (6)$$

All easy axes are rigidly connected to the particle, so their motion is described by the same angular velocity $\vec{\omega}$. In the overdamped (non-inertial) limit, this velocity can be obtained from the torque balance equation

$$\vec{\omega} = \frac{\mu_0}{\Gamma} [\vec{\mathcal{M}} \times \vec{H}] + \vec{\omega}^{\text{th}}. \quad (7)$$

Here, $\vec{H}$ is the *external* magnetic field acting on a multicore particle. $\vec{\mathcal{M}}$ is its total magnetic moment:

$$\vec{\mathcal{M}} = \sum_{i=1}^N \vec{\mu}_i. \quad (8)$$

Γ is the rotational friction coefficient, for a sphere in a viscous medium it is given by the Stokes' law

$$\Gamma = \pi\eta D^3. \quad (9)$$

Here, η is kinematic viscosity of a carrier liquid. The value of $\vec{\omega}^{\text{th}}$ is the random component of the angular velocity arising due to the particle Brownian motion, its autocorrelation function must satisfy the fluctuation-dissipation theorem:

$$\langle \omega_i^{\text{th}}(t) \omega_j^{\text{th}}(t') \rangle = \frac{1}{\tau_B} \delta_{ij} \delta(t - t'). \quad (10)$$

τ_B is the relaxation time of the MMNP rotational diffusion (*Brownian relaxation time*):

$$\tau_B = \frac{\Gamma}{2k_B T} = \frac{3\eta v_p}{k_B T}. \quad (11)$$

In the rotating reference frame rigidly connected to the MMNP, dynamics of the i -th magnetic moment is described by the standard Landau-Lifshitz-Gilbert equation. Neglecting fast Larmor precession and transitioning to the laboratory reference frame, one can write down this equation in the following form [closely mirroring Eqs. (6)-(7)]:

$$\frac{d\vec{e}_i}{dt} = \vec{\omega}_i^{(e)} \times \vec{e}_i, \quad (12)$$

$$\vec{\omega}_i^{(e)} = \vec{\omega} + \frac{\mu_0}{\Gamma_e} [\vec{\mu}_i \times \vec{h}_i^{(e)}] + \vec{\zeta}_i. \quad (13)$$

Here, $\vec{\omega}_i^{(e)}$ is the angular velocity of the i -th magnetic moment. Total effective magnetic field $\vec{h}_i^{(e)}$, acting on the moment, takes into account a combined effect of the external field, magnetic anisotropy and dipole-dipole interactions with other grains:

$$\vec{h}_i^{(e)} = \vec{H} - \frac{1}{\mu_0 \mu} \frac{\partial U_{\text{ani}}(i)}{\partial \vec{e}_i} - \frac{1}{\mu_0} \sum_{j \neq i} \frac{\partial U_{dd}(i, j)}{\partial \vec{\mu}_i}. \quad (14)$$

Γ_e is the “magnetic friction” coefficient:

$$\Gamma_e = \frac{\mu_0 \mu (1 + \alpha^2)}{\alpha \gamma}, \quad (15)$$

where α is the phenomenological Gilbert damping constant, γ is the gyromagnetic ratio. $\vec{\zeta}$ is the thermal noise that satisfies fluctuation-dissipation theorem:

$$\langle \zeta_{i,k}(t) \zeta_{j,l}(t') \rangle = \frac{1}{\tau_D} \delta_{ij} \delta_{kl} \delta(t - t'). \quad (16)$$

τ_D is the characteristic time scale of the magnetic moment rotational diffusion in the limit of zero anisotropy:

$$\tau_D = \frac{\Gamma_e}{2k_B T} = \frac{(1 + \alpha^2)\mu}{2\alpha\gamma k_B T}. \quad (17)$$

2.3. Dynamic magnetic susceptibility

The main focus of our work is the magnetic response of MMNPs to a weak probing oscillating field of frequency ω . The standard quantitative measure of such response is the dynamic magnetic susceptibility [92]:

$$\chi(\omega) = \frac{1}{v_p} \frac{\partial \langle \mathcal{M}_H \rangle}{\partial (H_0 e^{i\omega t})} \Big|_{H_0 \rightarrow 0}, \quad (18)$$

where $\mathcal{M}_H$ is the projection of the particle magnetic moment on the field direction, $\langle \dots \rangle$ denotes ensemble average, H_0 is the field magnitude. χ is a complex number, $\chi(\omega) = \chi'(\omega) - i\chi''(\omega)$.

Note, that in our notation χ characterizes the linear response on the level of a *single MMNP*. If one instead is interested in the susceptibility of a *suspension of MMNPs*, it is given by

$$\chi_{\text{susp}}(\omega) = \chi(\omega)\Phi, \quad (19)$$

where Φ is the volume fraction of multicore particles in the system. We do not account for interactions between different MMNPs in this work and the high-dilution limit $\Phi \ll 1$ is assumed.

2.3.1. Non-interacting grains

In the limiting case $\lambda \ll 1$, the rotation of magnetic moments inside the particle is entirely uncorrelated. We can expect, that the linear magnetic response of such system can be described by Shliomis-Stepanov approximation [87], which was initially proposed for the dilute suspension of single-core particles ($N = 1$):

$$\chi(\omega) = \frac{1}{3} \chi_{\parallel}(\omega) + \frac{2}{3} \chi_{\perp}(\omega). \quad (20)$$

This is a superposition of two components – $\chi_{\parallel}$ is the magnetic susceptibility along the particle easy axis and $\chi_{\perp}$ is the susceptibility in an orthogonal direction. They are both given by Debye-like expressions with different relaxation times $\tau_{\parallel}$ and $\tau_{\perp}$:

$$\chi_{\parallel}(\omega) = \frac{\chi_{\parallel}(0)}{1 + i\omega\tau_{\parallel}}, \quad (21)$$

$$\chi_{\perp}(\omega) = \frac{\chi_{\perp}(0)}{1 + i\omega\tau_{\perp}}. \quad (22)$$

Corresponding static (equilibrium) susceptibilities are given by

$$\chi_{\parallel}(0) = \chi_L(1 + 2G), \quad (23)$$

$$\chi_{\perp}(0) = \chi_L(1 - G), \quad (24)$$

where χ_L is the so-called Langevin susceptibility – the equilibrium susceptibility of an ideal gas of magnetically isotropic particles:

$$\chi_L = \frac{\mu_0 \mu^2 N}{3k_B T v_p}. \quad (25)$$

The order parameter G defines the average degree of alignment between the particle magnetic moment and its easy axis in the absence of the field. It can be calculated as [91]

$$G = \frac{3}{2} \langle (\vec{e} \cdot \vec{n})^2 \rangle_{H=0} - \frac{1}{2} \quad (26)$$

$$= \frac{3}{4} \frac{1}{\kappa} \left[\frac{\sqrt{\kappa}}{\text{Da}(\sqrt{\kappa})} - 1 \right] - \frac{1}{2}, \quad (27)$$

where

$$\text{Da}(x) = e^{-x^2} \int_0^x e^{y^2} dy$$

is the so-called Dawson integral. Relaxation times are given by

$$\tau_{\parallel} = \left(\frac{1}{\tau_{\parallel}^N} + \frac{1}{\tau_B} \right)^{-1}, \quad (28)$$

$$\tau_{\perp} = \left(\frac{1}{\tau_{\perp}^N} + \frac{1}{\tau_B} \right)^{-1}, \quad (29)$$

where time scales $\tau_{\parallel}^N$ and $\tau_{\perp}^N$ describe the so-called *Néel relaxation* of magnetic moments in the absence of mechanical (Brownian) rotation of the particle. Specifically, $\tau_{\parallel}^N$ corresponds to rare spontaneous jumps of the magnetic moment between two energetically equivalent states allowed by the uniaxial anisotropy (“*interwell*” relaxation in Raikher-Stepanov terminology [93]). For it, we will use a well-established approximation by Cregg et al. [94]:

$$\tau_{\parallel}^N = \tau_D \frac{e^{\kappa} - 1}{2\kappa} \left[\frac{1}{1 + 1/\kappa} \sqrt{\frac{\kappa}{\pi}} + 2^{-\kappa-1} \right]^{-1}. \quad (30)$$

The second time $\tau_{\perp}^N$ describes fast fluctuations of the magnetic moment while in one of two equilibrium states (“*intrawell*” relaxation). It should be well approximated by the formula [87,95]

$$\tau_{\perp}^N = 2\tau_D \frac{1 - G}{2 + G}. \quad (31)$$

2.3.2. Equilibrium susceptibility

In Refs. [72,73], we introduced and tested the following formula for the equilibrium ($\omega = 0$) susceptibility of a multicore-based ferrofluid with magnetically isotropic grains ($\kappa \ll 1$):

$$\chi_{\text{susp}}(0) = \chi_L \Phi \frac{1 + \chi_L/3}{1 + \chi_L(1 + \chi_L/3)(1 - \Phi)/3}. \quad (32)$$

In the infinite dilution limit, Eq. (32) reduces to the following equation for a static susceptibility of an isolated MMNP:

$$\chi(0) = \chi_L \frac{1 + \chi_L/3}{1 + \chi_L(1 + \chi_L/3)/3}. \quad (33)$$

The deviation of $\chi(0)$ from χ_L in this expression illustrates the effect of intergrain dipole-dipole interactions on the static magnetic response of the MMNP.

2.3.3. Specific loss power

If the susceptibility is known, one can use its imaginary part to estimate the heating ability of MMNPs, which is relevant for magnetic hyperthermia applications. In a harmonically oscillating field, the amount of energy dissipated into heat per unit of time per unit of suspension volume is given by [96]

$$Q = \frac{\mu_0 H_0^2}{2} \omega \chi''_{\text{susp}}(\omega). \quad (34)$$

Instead of the volumetric dissipation, practitioners often focus on the “specific loss power” (or “specific absorption rate”), which measures the heating rate per unit of mass of the magnetic material [93,97]:

$$\text{SLP} = \frac{Q}{\rho}, \quad (35)$$

where ρ is the mass density of nanoparticles (grains in our case) in the suspension.

In order to isolate effects of intergrain interactions and anisotropy on the heating ability of MMNPs at different frequencies, let us introduce the following reduced form of the loss power:

$$q = \frac{2\rho\tau_D}{\mu_0 H_0^2 \Phi} \text{SLP} = \omega\tau_D \chi''(\omega). \quad (36)$$

For non-interacting grains with zero anisotropy, the Shliomis-Stepanov approximation (Sec. 2.3.1) predicts that in a high-frequency limit SLP must reach a constant value:

$$q(\omega\tau_D \gg 1) = \chi_L. \quad (37)$$

2.4. Brownian dynamics simulations

To calculate dynamic susceptibility at arbitrary values of κ and λ , we employed the Brownian dynamics simulation method. All calculations were performed using software package ESPResSo 4.2 [98]. The Egg model (Sec. 2.2) was implemented on the basis of the existing Brownian dynamics feature of ESPResSo, which uses Ermak/McCammon propagation scheme [99]. ESPResSo “virtual sites” feature was used to create a rigid coupling between easy axes and the MMNP body.

Initial susceptibilities were calculated from fluctuations of the MMNP magnetic moment in zero field [100]. Specifically, static susceptibility was calculated as

$$\chi(0) = \chi_L \frac{\langle \vec{M}^2 \rangle}{\mu^2 N}, \quad (38)$$

and the normalized dynamic susceptibility was estimated as

$$\frac{\chi(\omega)}{\chi(0)} = 1 + i\omega \int_0^\infty C_M(t) e^{i\omega t} dt \quad (39)$$

where $C_M(t)$ is the magnetic moment autocorrelation function

$$C_M(t) = \frac{\langle \vec{M}(t) \cdot \vec{M}(0) \rangle}{\langle \vec{M}^2 \rangle}. \quad (40)$$

In practice, to perform Fourier transform in Eq. (39), calculated autocorrelation functions were first approximated with a sum of exponents

$$C_M(t) = \sum_{j=1}^n f_j e^{-t/\tau_j} \quad (41)$$

with the constrain $\sum_{j=1}^n f_j = 1$. No more than four exponents had to be used. Dipole-dipole interactions between each pair of grains were calculated directly without truncation.

Simulations were performed using reduced time $t^* = t/\tau_D$. The time step was equal to $\Delta t^* = 0.005$, and the ratio of Brownian to magnetic time scales was set to $\tau_B/\tau_D = 200$. In each simulation, first 2000 time steps were used to equilibrate the system, and subsequent 4×10^7 time steps were used for data collection. For each type of magnetic texture and each values of λ and κ , five different realizations of MMNP were created and simulated independently. The final results are averaged over these five realizations.

3. Results and discussions

First, we investigate the interplay between magnetic anisotropy of the cores and magnetic intercore interactions. For that, we analyse how the anisotropy field keeps the dipoles aligned with the anisotropy axes.

Fig. 2 illustrates histograms of the orientations of magnetic moments in the cores of multicore particles, where all core anisotropy axes are aligned parallel. The horizontal axis, $\cos^2 \alpha$, represents the squared cosine of the angle α between a magnetic moment of the core and the easy axis, while the vertical axis shows the fraction of cores that adopt a given orientation. The three panels, labelled Fig. 2(a), 2(b), and 2(c), correspond to increasing values of the magnetic interaction strength, λ , which characterises the dipole-dipole interaction between the cores. Specifically, the panels represent $\lambda = 1, 5$, and 10, respectively. Within each panel, three data sets illustrate the influence of different values of the anisotropy constant, κ , which measures the strength of the magnetic anisotropy of individual cores, namely $\kappa = 1, 5$, and 10.

In Fig. 2(a), corresponding to $\lambda = 1$, the orientation distribution is primarily determined by the anisotropy constant κ . For low anisotropy, $\kappa = 1$, the distribution is relatively flat, indicating weak alignment of magnetic moments with the easy axis. As κ increases to 5 and 10, a significant shift is observed, with magnetic moments becoming increasingly concentrated near $\cos^2 \alpha = 1$, reflecting the stronger alignment induced by the anisotropy.

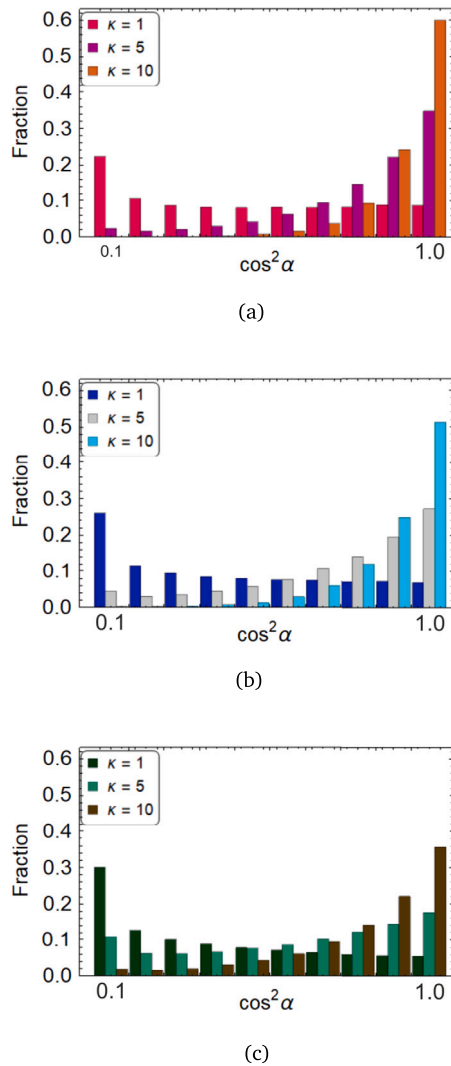


Fig. 2. Histograms of magnetic moment orientations, $\cos^2 \alpha$, in multicore particles with all core anisotropy axes aligned parallel. Each panel shows the distribution for different anisotropy constants, κ , as indicated by colours in the legends. The panels show magnetic moment orientations for different values of λ : (a) $\lambda = 1$, (b) $\lambda = 5$, and (c) $\lambda = 10$.

Fig. 2(b), where $\lambda = 5$, demonstrates the effects of intermediate interaction strength. Here, the interplay between anisotropy and intercore interactions becomes evident. For low anisotropy ($\kappa = 1$), the magnetic interactions dominate, preventing significant alignment of the magnetic moments along the easy axis, as shown by the distribution with a small maximum at the orthogonal orientation. In fact, parallel orientation of strongly interacting dipoles is unlikely. However, as κ increases to 5 and 10, the distributions exhibit a sharper peak near $\cos^2 \alpha = 1$.

In Fig. 2(c), corresponding to $\lambda = 10$, the interaction strength is sufficiently high for collective effects to dominate. The distributions across all values of κ are no longer concentrated near $\cos^2 \alpha = 1$. Unless $\kappa = 10$, the influence of individual anisotropy becomes less pronounced as the magnetic moments are primarily governed by collective interactions.

These results reveal that the alignment of magnetic moments in multicore particles is governed by the interplay between the anisotropy constant, κ , and the magnetic interaction strength, λ . At low interaction strength, the anisotropy effects dominate, leading to a gradual increase in alignment as κ increases. However, as λ grows, intercore interactions progressively lead to the energetic disadvantage of moment parallel orientation, eventually overshadowing the effects of anisotropy at high values of λ . The histograms provide a clear representation

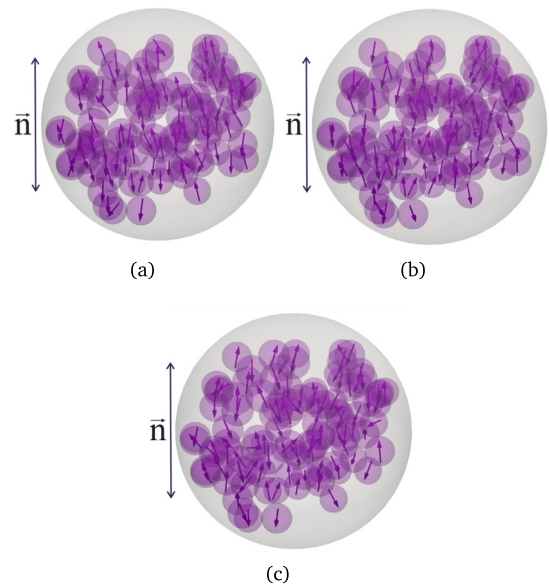


Fig. 3. Simulation snapshots for $\kappa = 10$. The panel shows magnetic moment orientations of grains for different values of λ : (a) $\lambda = 1$, (b) $\lambda = 5$, and (c) $\lambda = 10$.

of the transition from anisotropy-dominated to interaction-dominated regimes, highlighting the cooperative nature of these effects.

The evolution of the snapshots from Fig. 3(a) to 3(c) demonstrates the impact of increasing interaction strength between magnetic grains on their magnetic moment orientations, while the anisotropy remains constant and the same for all cores, as shown by a double-sided arrow. For low interaction strength ($\lambda = 1$), as shown in the snapshot Fig. 3(a), the system exhibits a very high alignment with the easy axis. In this case the anisotropy energy is much higher than the dipolar interactions.

As the interaction strength increases to $\lambda = 5$, shown in snapshot Fig. 3(b), the magnetic moments begin to show noticeable correlations. Some regions, particularly in the centre of the multicore, display tendencies toward mutual alignment or the formation of local structures, indicating that the dipolar interactions are becoming significant compared to thermal fluctuations. Less aligned are the particles near the periphery.

In snapshot Fig. 3(c), where $\lambda = 10$, the interaction strength strongly affects system behaviour. One can see several magnetic moments aligned clearly perpendicular to the easy axis. These arrangements arise as the system seeks to minimize its total energy under the strong influence of dipolar forces. The anisotropy is not anymore strong enough to keep all the core dipoles aligned.

The aforementioned interplay cannot but manifest itself in the zero-field dynamic susceptibility, as shown in Figs. 4.

Fig. 4 presents the dynamic magnetic susceptibility spectra, normalized as $\chi(\omega)/\chi(0)$, for multicore particles with parallel anisotropy axes, under varying conditions of anisotropy constant κ and intercore interaction strength λ . The horizontal axis represents the dimensionless frequency $\omega\tau_D$, where $\omega\tau_D = 1$ corresponds to the characteristic rotational frequency of an isolated core. The panels (a), (b), and (c) show the system behaviour for increasing interaction strengths $\lambda = 1, 5$, and 10, respectively, while different colours within each panel distinguish the spectra for anisotropy constants $\kappa = 1, 5$, and 10. Both the real (χ') and imaginary (χ'') components of the susceptibility are displayed.

For Fig. 4(a) ($\lambda = 1$), the system is in the weak interaction regime, where the cores largely behave independently. The real part χ' is nearly flat at low frequencies, while the imaginary part χ'' features a single relaxation peak around $\omega\tau_D \approx 1$, corresponding to basically Debye-like behaviour. As κ increases, the relaxation peak splits: one part shifts to lower frequencies, underlining the increasing role of the multicore particle rotation, whose characteristic frequency is $\omega\tau_B^{mult} = 0.005$; and the

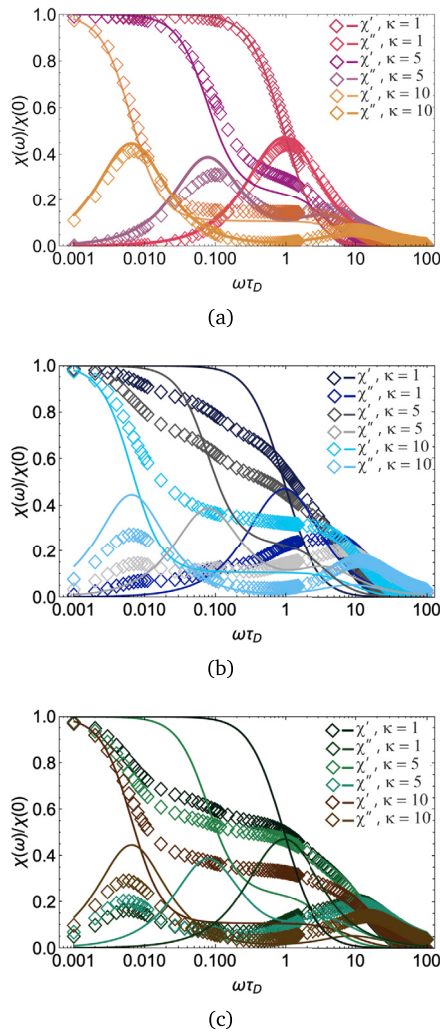


Fig. 4. Dynamic susceptibility spectra, $\chi(\omega)/\chi(0)$, for multicore particles with parallel anisotropy axes, as a function of the dimensionless frequency $\omega\tau_D$ and varying anisotropy constant κ and intercore interaction strength λ . The panels show the spectra for different values of λ : (a) $\lambda = 1$, (b) $\lambda = 5$, and (c) $\lambda = 10$. The spectra are plotted for various values of $\kappa = 1, 5, 10$, with the real (χ') and imaginary (χ'') parts of the susceptibility represented. $\omega\tau_D = 1$ corresponds to the eigen frequency of a single-core Néel rotation in the absence of interactions. The solid lines correspond to the theoretical predictions for non-interacting cores Eq. (20)–(31).

second peak moves towards higher frequencies, reflecting the increasing difficulty of reorienting magnetic moments due to stronger anisotropy. For small values of λ , the interactions between cores are minimal, resulting in spectra that closely resemble the behaviour of individual cores. In such cases, the theoretical predictions for non-interacting cores given by Eq. (20)–(31), shown as solid lines, closely align with the numerical results.

Fig. 4(b) ($\lambda = 5$) represents intermediate interaction strengths, where competition between anisotropy and intercore interactions gives rise to more complex dynamics. For low κ , interactions dominate, leading to broader and slightly shifted peaks in χ'' , indicating a distribution of relaxation times resulting from intercore coupling. In this case, the role of the multicore rotation as a whole is less important. At higher κ , anisotropy becomes significant and limits the influence of interactions, narrowing and lowering the high-frequency peaks and pushing the relaxation frequencies away from those characteristic of single-core behaviour. In general, what we see here is that the interactions lead to the decrease of the low-frequency peak, if compared to Fig. 4(a) ($\lambda = 1$), while anisotropy growth promotes the overall multicore relaxation.

Fig. 4(c) ($\lambda = 10$) shows the strong interaction regime, and confirm the aforementioned interplay between intercore interactions and anisotropy fields. Here, intercore coupling dictates the system dynamics. The spectra reveal significant deviations from single-core relaxation behaviour. Peaks in χ'' broaden and exhibit complex shapes, reflecting collective modes driven by strong interactions. For all κ values, the relaxation dynamics become increasingly cooperative, and the individual effects of anisotropy are partially masked. Nevertheless, as κ increases, anisotropy can still influence relaxation times, causing minor shifts in the peak position and changes in peak broadening.

Overall, the figure demonstrates how the dynamic susceptibility spectra are “polluted” or altered because of the interplay of anisotropy and intercore interactions. In the weak-interaction regime and low anisotropy energies, the spectra closely resemble those of independent cores. As λ grows, collective effects dominate; particularly at low anisotropy, the high-frequency peaks broaden, while the low-frequency peak lowers and narrows. Note that here, for large values of λ , a parallel pattern of the easy axis leads to a very specific type of orientational correlation, as the parallel alignment of side-by-side dipoles is energetically disadvantageous. At higher κ , anisotropy can still impose its characteristic relaxation dynamics, emphasizing the overall multicore low-frequency relaxation (compare the positions of the low-frequency peaks for $\kappa = 10$ in all three sub-figures).

As it is not easy to create multicores with a parallel patterning of easy axes, it is also important to understand what happens when the axes are randomised.

The dynamic susceptibility spectra of multicore systems with randomly oriented anisotropy axes are presented in Fig. 5 for different values of κ and λ . Compared to the case of parallel anisotropy axes, the random orientation introduces additional complexities in the relaxation dynamics, that however basically do not manifest themselves in χ' and χ'' components.

For low λ values, intercore interactions are weak, and the spectra closely resemble the single-core behaviour. In these cases, the theoretical predictions for non-interacting cores Eq. (20)–(31), represented by solid lines, match the numerical results well. However, as λ increases, intercore interactions begin to influence the relaxation dynamics significantly. The peak at $\omega\tau_D = 1$ diminishes, and additional features emerge in the low-frequency region due to the collective effects of interactions between cores. For large values of λ , the spectra are highly broadened, and the dynamic response becomes dominated by the interplay between anisotropy and intercore coupling, resulting in complex multi-mode relaxation behaviour, similar to the ones discussed above.

In general, independently of the easy-axes distribution, for high κ , the spectra reflect the dominance of Brownian motion of the multicore particle as a whole. The effect of increasing λ leads to an effective increase of the anisotropy of interacting cores, so the low-frequency peak is even further enhanced.

In order to strengthen the conclusions we obtained from the dynamic susceptibility spectra, and get a closer look at the impact of core-axes distributions, we present Fig. 6.

The plots in Fig. 6 illustrate the normalised initial static susceptibility $\chi(0)/\chi_L$ for multicore systems, comparing parallel, Fig. 6(a) and random Fig. 6(b) distributions of easy axes as a function of λ . The normalisation by the Langevin susceptibility χ_L for $\lambda = 0$ provides insight into the relative suppression of susceptibility due to dipolar interactions and anisotropy effects.

Despite the normalized susceptibility $\chi(0)/\chi_L$ decreasing with λ , it is notable that the non-normalized susceptibility $\chi(0)$ actually grows with λ as shown in Table 1.

This simultaneous growth and suppression reveal that the dipolar forces remain significant, strongly influencing the system collective behaviour. While the raw increase in $\chi(0)$ reflects the enhanced response of the cores to intercore interactions, the normalized suppression indicates that these forces suppress correlations between the dipoles, hindering a uniform response to the external field. This highlights the

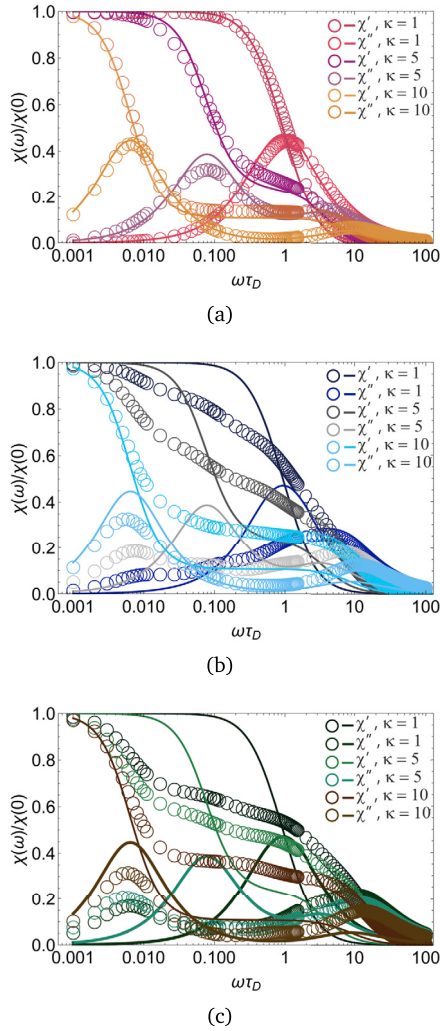


Fig. 5. Dynamic susceptibility spectra $\chi'(\omega)/\chi(0)$ and $\chi''(\omega)/\chi(0)$ of multicore systems with randomly oriented anisotropy axes for varying κ and λ : (a) $\lambda = 1$, (b) $\lambda = 5$, (c) $\lambda = 10$. The x-axis represents the reduced frequency $\omega\tau_D$, where τ_D is the characteristic relaxation time of a single core. The solid lines correspond to the theoretical predictions for non-interacting cores Eq. (20)–(31).

Table 1

Values of static zero-field magnetic susceptibility, $\chi(0)$ for $\lambda = 1, 5, 10$ as stated in the first row, and $\kappa = 1, 5, 10$ (first column). The last row contains corresponding values of Langevin susceptibility, χ_L . For each pair $\lambda - \kappa$, two values of $\chi(0)$ are given: for Random mutual orientation of easy axes in a multicore and for the Parallel one.

λ	1		5		10	
κ	R	P	R	P	R	P
1	1.332	1.322	2.669	2.656	3.050	3.193
5	1.315	1.157	2.858	2.477	3.414	3.361
10	1.273	1.118	2.948	2.043	4.031	3.581
χ_L	1.6		8		16	

complex interplay between dipolar interactions, anisotropy, and spatial constraints, which prevents the dipoles from aligning freely and leads to vortex-like or ring-like configurations that reduce the system's net response.

Another important observation is the role of $\kappa = 10$, which introduces deviations between parallel and random configurations at higher values of λ . For $\kappa = 10$, the strong anisotropy in the parallel case results in a more coherent alignment of the dipoles along the shared easy

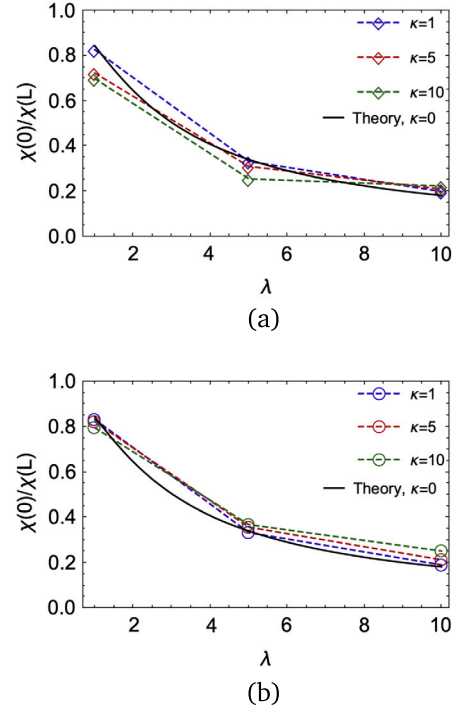


Fig. 6. Initial static susceptibility $\chi(0)$ normalized by Langevin susceptibility χ_L as a function of λ for multicores with parallel (a) and random (b) easy axis distributions. Solid lines represent mean-field theoretical predictions for $\kappa = 0$, Eq. (33) [72,73]. Symbols are simulation data. Dashed lines, connecting the points are guides for the eye. Values of κ are reflected in the colour, as indicated in the legend.

axis, delaying the onset of significant suppression in $\chi(0)/\chi_L$ compared to the random case. In contrast, the random distribution creates a more isotropic energy landscape, where dipolar interactions and anisotropy effects compete more uniformly, leading to a more pronounced suppression of susceptibility. These deviations underscore the sensitivity of the system response to the alignment of anisotropy axes and demonstrate the intricate role of κ in shaping the magnetic behaviour of multicore systems.

The solid lines in the plots represent theoretical predictions that agree well with the simulation data, particularly for weaker interactions.

Finally, the susceptibility $\chi(0)$ does not go to zero for growing λ , as the one for similar cores in a suspension would [101]. Fixed spatial arrangements of the cores prevent the dipoles from assuming energy-minimizing configurations with nearly zero magnetisation. So, the multicores remain more responsive compared to systems with mobile cores, but the impact of interactions and anisotropy still significantly decreases the susceptibility if compared to non-interacting cores.

As the last step, we calculated q from Eq. (36) normalised by χ_L , as the main contribution to SLP, Eq. (35). The results below are shown for multicores with random distribution of easy axes only, as the results for the case of easy axes being parallel to each other are indistinguishable. The panels correspond to different interaction strengths λ : Fig. 7(a) $\lambda = 1$, Fig. 7(b) $\lambda = 5$, and Fig. 7(c) $\lambda = 10$. The horizontal axis is the dimensionless frequency $\omega\tau_D$, where τ_D is the characteristic Brownian relaxation time for a single grain.

In Fig. 7(a), where the interaction strength is weak ($\lambda = 1$), for small anisotropy the SLP curve exhibits a single distinct hunch around $\omega\tau_D \sim 1$, corresponding to the optimal frequency where eigen core relaxation dominates energy dissipation. As κ increases to 5, the splits in two (one at a lower than $\omega\tau_D \sim 1$ frequency and one at a higher), indicating reduced dissipation efficiency due to the moderate anisotropy. For $\kappa = 10$, the hunches are well separated by a plateau, suggesting that high anisotropy suppresses energy absorption efficiency.

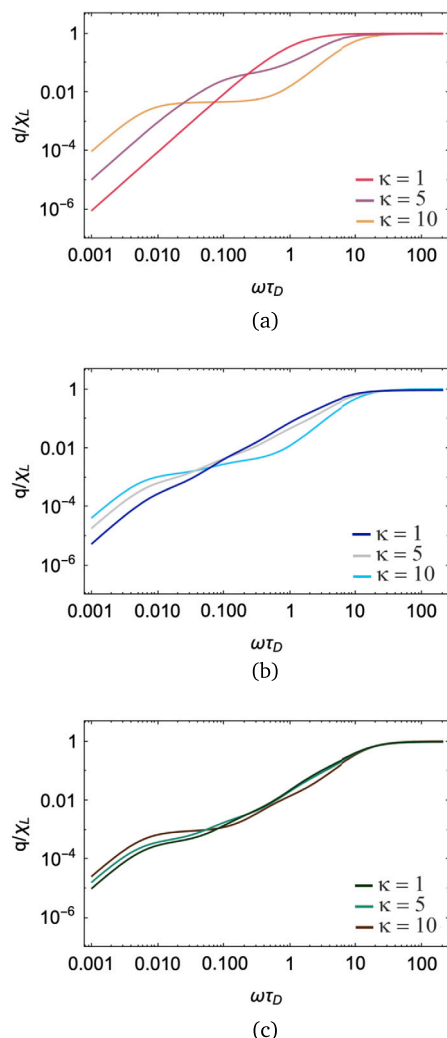


Fig. 7. Reduced form of the loss power, q , from Eq. (36): (a) $\lambda = 1$, (b) $\lambda = 5$, (c) $\lambda = 10$. The x-axis represents the reduced frequency $\omega\tau_D$, where τ_D is the characteristic relaxation time of a single core.

In Fig. 7(b), where the interaction strength is moderate ($\lambda = 5$), dipolar interactions begin to influence the dissipation dynamics. For $\kappa = 1$, the hunch is single, but shifts slightly toward higher frequencies, reflecting the onset of interaction effects. As κ increases to 5, the second hunch at lower frequencies arises, but the amplitude decreases compared to Fig. 7(a). At $\kappa = 10$, the hunch shifts even more to the left, indicating the combined effects of stronger anisotropy and interactions on energy dissipation.

In Fig. 7(c), at strong interaction strength ($\lambda = 10$), dipolar interactions dominate the system behaviour. For $\kappa = 1$, the peak broadens considerably and shifts to lower frequencies compared to the weaker interaction cases. All three curves show two hunches with amplitudes lower.

In general, the results demonstrate that increasing the strength of the interaction λ and the anisotropy κ suppress the specific loss power and split/shift the dissipation hunches to lower and higher frequencies, than the eigen individual core ones. This highlights the intricate interplay between magnetic anisotropy and dipolar interactions in determining the dynamics of the system.

4. Conclusions

In this study, we employed Brownian dynamics simulations to investigate the static and dynamic responses of magnetic multicore nanopar-

ticles, focusing on the effects of core magnetic anisotropy and intercore interaction strength.

At low interaction strength ($\lambda = 1$, comparable to thermal fluctuations), magnetic anisotropy (κ) dominates, aligning magnetic moments with the easy axes. At intermediate interaction strengths ($\lambda = 5$), dipolar interactions become significant, suppressing alignment for low anisotropy while maintaining it at higher κ . For strong intercore coupling ($\lambda = 10$), collective dipolar effects prevail, leading to broader magnetic moment distributions and reduced alignment, except at very high anisotropy ($\kappa = 10$).

The dynamic susceptibility spectra reflect a transition from independent core-like behaviour at weak interactions to collective dynamics at higher λ . Strong intercore interactions at low κ produce broader, shifted peaks, while higher κ imposes a distinct low-frequency relaxation pattern across all interaction regimes.

Notably, we found no significant effect of easy-axes texturing on the spectra.

Our results show that increasing both interaction strength (λ) and anisotropy (κ) reduces specific loss power and shifts dissipation hunches to lower frequencies. This underscores the intricate interplay between anisotropy and dipolar interactions in shaping the magnetic and dynamic properties of multicore systems.

These findings provide valuable insights into the magnetic behaviour of multicore nanoparticles and highlight the importance of anisotropy and interaction strength in tailoring their properties. This knowledge is essential for advancing applications such as magnetic hyperthermia, data storage, and targeted drug delivery with both passive and functionalised active magnetic colloids, where precise control of magnetic dynamics is crucial.

CRediT authorship contribution statement

Elena S. Pyanzina: Writing – review & editing, Validation, Formal analysis. **Ekaterina V. Novak:** Methodology, Data curation, Conceptualization. **Andrey A. Kuznetsov:** Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Sofia S. Kantorovich:** Writing – review & editing, Writing – original draft, Project administration, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Elena Pyanzina reports financial support was provided by Russian Science Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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